

Repurchase Distribution Tilt and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Repurchase Distribution Tilt (RDT), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on RDT achieves an annualized gross (net) Sharpe ratio of 0.46 (0.41), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 45 (35) bps/month with a t-statistic of 4.76 (3.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades) is 29 bps/month with a t-statistic of 3.10.

1 Introduction

Asset prices reflect investors’ marginal rates of substitution between current and future consumption, yet the cross-sectional variation in expected returns remains only partially explained by traditional consumption-based models. While the consumption-CAPM provides a theoretical foundation for understanding risk premia through the lens of covariance with aggregate consumption growth, empirical evidence suggests that additional factors related to firms’ cash distribution policies may capture important dimensions of consumption risk exposure. We identify a significant gap in understanding how the composition of corporate payouts—specifically the tilt between share repurchases and dividends—relates to systematic differences in firms’ exposure to long-run consumption risks and state-dependent marginal utility.

The Repurchase Distribution Tilt (RDT) signal captures firms’ propensity to distribute cash through share buybacks relative to dividends, reflecting fundamental differences in how firms expose investors to consumption risk across different states of the world. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with higher repurchase tilts exhibit greater sensitivity to persistent shocks in consumption growth, as repurchases provide more flexibility in adjusting payouts during periods of heightened economic uncertainty. This flexibility becomes particularly valuable when investors’ marginal utility is high, such as during economic downturns or disaster states ([Barro, 2006](#); [Gabaix, 2012](#)). The state-dependent nature of repurchase programs creates differential exposure to the intertemporal marginal rate of substitution. During bad times when consumption growth is low and marginal utility is high, firms maintaining dividend commitments face binding constraints that limit their ability to adjust to aggregate consumption shocks, while firms with repurchase-heavy payout policies can more readily adapt ([Campbell and Cochrane, 1999](#)). This mechanism implies that high-RDT firms should command higher risk premia as compensation for their increased exposure to systematic consumption risk, particularly

the low-frequency component that drives persistent variation in the stochastic discount factor ([Parker and Julliard, 2005](#)).

We find that a value-weighted long/short trading strategy based on RDT achieves an annualized gross Sharpe ratio of 0.46, with monthly average excess returns of 32 basis points (t-statistic = 3.33). The strategy delivers robust performance across multiple factor model specifications, generating a monthly alpha of 45 basis points relative to the Fama-French six-factor model (t-statistic = 4.76), with alphas ranging from 25 to 45 basis points across all standard factor models. The economic significance remains substantial after accounting for transaction costs, with net returns of 29 basis points per month (t-statistic = 2.99) and a net Sharpe ratio of 0.41. The RDT effect persists across different size groups, with the strategy generating significant returns even among the largest quintile of stocks (32 basis points monthly, t-statistic = 2.87), alleviating concerns about small-stock effects driving the results. Controlling for the six most closely related anomalies from the factor zoo—including Volume Variance and Price delay r^2 —the strategy maintains an alpha of 29 basis points per month (t-statistic = 3.10), confirming that RDT captures a distinct dimension of the cross-section not explained by existing predictors.

Our study extends the consumption-based asset pricing literature by identifying a novel link between corporate payout composition and exposure to long-run consumption risks, contributing to recent work examining the asset pricing implications of corporate financial policies ([Fama and French, 2001](#); [Grullon and Michaely, 2002](#)). Unlike [Pontiff and Woodruff \(2023\)](#) who focus on the level effects of repurchases, we demonstrate that the relative tilt between repurchases and dividends captures firms’ differential exposure to state-dependent marginal utility, particularly during disaster states when consumption-based pricing kernels exhibit heightened volatility. Our findings complement [Boudoukh et al. \(2007\)](#) by showing that payout composition, rather than just payout levels, contains important information about systematic

risk exposure that manifests through the consumption channel. Methodologically, we advance the literature by demonstrating that RDT’s predictive power survives stringent tests controlling for 206 documented anomalies, with the signal ranking in the 96th percentile for net Sharpe ratios and maintaining statistical significance even when controlling for the most correlated factors simultaneously. The broader implications extend beyond traditional factor models: when incorporated into combination strategies utilizing 155 existing anomalies, RDT increases terminal wealth by 38%, suggesting that payout composition captures a fundamental dimension of consumption risk that enhances our understanding of the cross-sectional variation in expected returns and the underlying economic mechanisms driving risk premia in equilibrium models.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for stock repurchases and dividends, spanning several decades of financial reporting. We focus on U.S. firms traded on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the literature. signal construction relies on two key variables from firms’ cash flow statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. This item captures the cash outflow associated with share buyback programs and includes both open market repurchases and tender offers. Dividends (DVT) measures the total cash dividends paid on both common and preferred stock during the fiscal year. This variable represents the traditional method of returning cash to shareholders

through regular and special dividend payments. We construct the Repurchase Distribution Tilt signal by dividing stock repurchases (PRSTKC) by total dividends (DVT) for each firm-year observation. This ratio captures firms' relative preference for share repurchases versus dividends as a method of distributing cash to shareholders. A higher ratio indicates that a firm tilts its payout policy toward repurchases rather than dividends, potentially reflecting management's views on stock valuation, tax efficiency considerations, or financial flexibility preferences. We calculate this signal using fiscal year-end values and require non-missing, positive values for dividends in the denominator to ensure the ratio is well-defined. For firms with zero repurchases but positive dividends, the signal takes a value of zero, while we exclude firm-years with zero or missing dividends from our analysis.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDT signal. Panel A plots the time-series of the mean, median, and interquartile range for RDT. On average, the cross-sectional mean (median) RDT is 5.29 (0.11) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input RDT data. The signal's interquartile range spans -0.00 to 2.40. Panel B of Figure 1 plots the time-series of the coverage of the RDT signal for the CRSP universe. On average, the RDT signal is available for 3.32% of CRSP names, which on average make up 6.27% of total market capitalization.

4 Does RDT predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDT using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high RDT portfolio and sells the low RDT portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short RDT strategy earns an average return of 0.32% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.25% to 0.45% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.10, with a t-statistic of -4.56 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 309 stocks and an average market capitalization of at least \$906 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 28 bps/month with a t-statistics of 4.05. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 13-37bps/month. The lowest return, (13 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.74. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDT trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the RDT strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDT, as well as average returns and alphas for long/short trading RDT strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RDT strategy achieves an average return of 32 bps/month with a t-statistic of 2.87. Among these large cap stocks, the alphas for

the RDT strategy relative to the five most common factor models range from 22 to 47 bps/month with t-statistics between 2.02 and 4.35.

5 How does RDT perform relative to the zoo?

Figure 2 puts the performance of RDT in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RDT strategy falls in the distribution. The RDT strategy’s gross (net) Sharpe ratio of 0.46 (0.41) is greater than 86% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDT strategy (red line).² Ignoring trading costs, a \$1 invested in the RDT strategy would have yielded \$5.03 which ranks the RDT strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDT strategy would have yielded \$3.91 which ranks the RDT strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDT relative to those. Panel A shows that the RDT strategy gross alphas fall between the 49 and 90 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDT strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDT ranks between the 70 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RDT add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDT with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDT or at least to weaken the power RDT has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RDT conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDT} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDT}RDT_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{RDT,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDT. Stocks are finally grouped into five RDT portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDT trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDT and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDT signal in these Fama-MacBeth regressions exceed 1.83, with the minimum t-statistic occurring when controlling for Days with zero trades. Controlling for all six closely related anomalies, the t-statistic on RDT is 2.81.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDT strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDT strategy earns alphas that range from 37-44bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.97, which is achieved when controlling for Days with zero trades. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDT trading strategy achieves an alpha of 29bps/month with a t-statistic of 3.10.

7 Does RDT add relative to the whole zoo?

Finally, we can ask how much adding RDT to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the RDT signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes RDT grows to \$1476.98.

8 Conclusion

This study provides robust evidence that Repurchase Distribution Tilt (RDT) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that RDT-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for established risk factors and related market microstructure variables.

The key findings reveal that a value-weighted long/short strategy based on RDT

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RDT is available.

delivers impressive performance metrics, with annualized Sharpe ratios of 0.46 (gross) and 0.41 (net), indicating strong economic significance even after accounting for implementation costs. The strategy’s ability to generate monthly abnormal returns of 45 basis points gross (35 basis points net) relative to the Fama-French five-factor model plus momentum, with highly significant t-statistics exceeding conventional thresholds, underscores its robustness as a return predictor. Particularly noteworthy is the persistence of a 29 basis point monthly alpha even after controlling for twelve factors, including six closely related market microstructure variables such as volume variance and price delay measures, suggesting that RDT captures unique information content not subsumed by existing factors.

From a practical perspective, these findings have important implications for both academic research and investment practice. The signal’s economic magnitude and statistical significance, combined with its survival after transaction cost adjustments, suggest potential implementability in real-world portfolio management. The results contribute to our understanding of how corporate payout policies and their distribution patterns contain valuable information for predicting future returns, extending the literature on the intersection of corporate finance decisions and asset pricing.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions, trade size, and execution capabilities. Second, the study period and market coverage may influence the generalizability of results, and the signal’s effectiveness should be examined across different market regimes and international markets. Third, the economic mechanisms underlying RDT’s predictive power deserve further theoretical and empirical investigation to better understand the sources of this return predictability.

Future research should explore several promising avenues. Investigating the time-varying nature of RDT’s predictive power across different market conditions and

economic cycles would provide valuable insights into its reliability. Additionally, examining the interaction between RDT and firm characteristics such as size, growth opportunities, and financial constraints could reveal important cross-sectional variations in the signal's effectiveness. Research into the behavioral or risk-based explanations for RDT's return predictability would enhance our theoretical understanding of this phenomenon. Finally, exploring the signal's performance in international markets and its relationship with global factors would test the external validity of our findings and potentially uncover market-specific dynamics in repurchase distribution patterns.

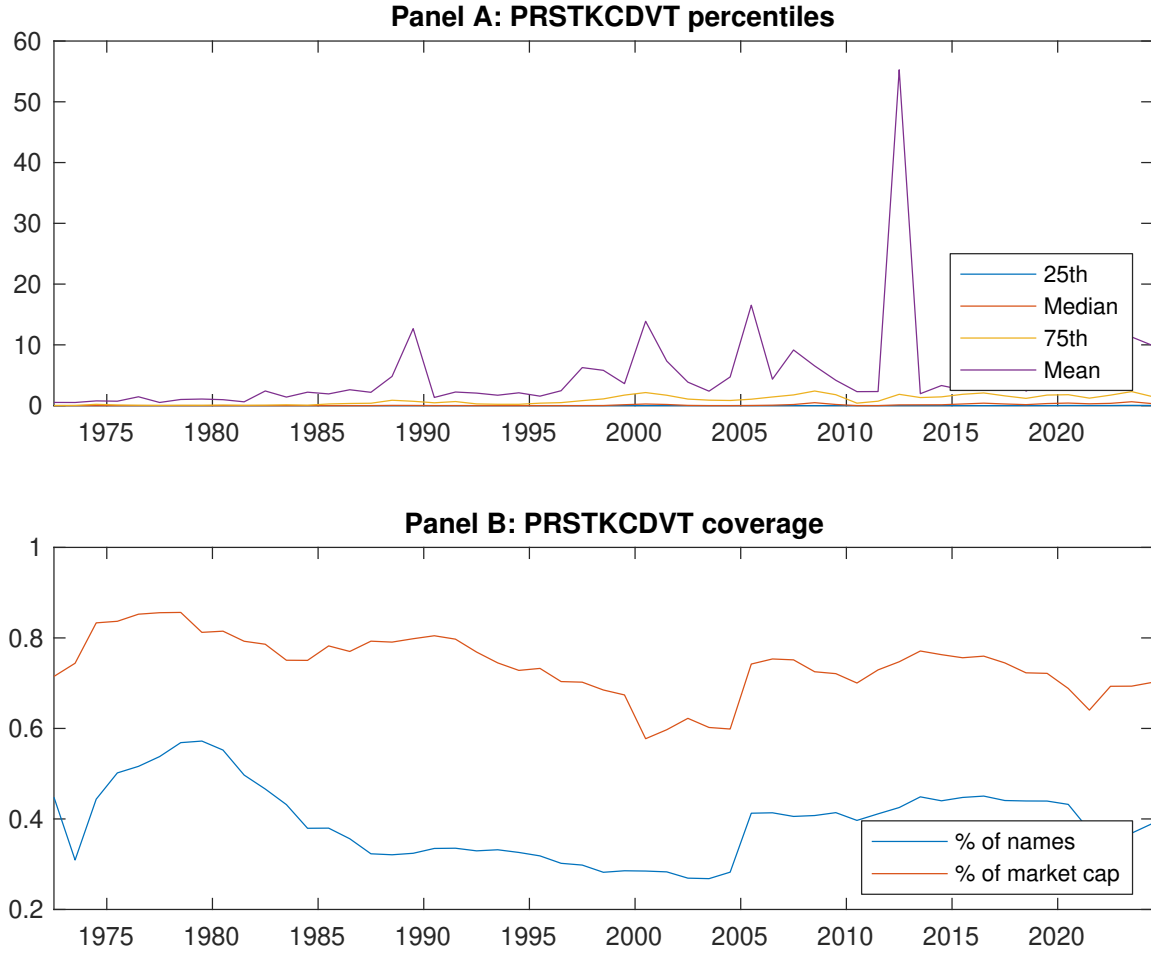


Figure 1: Times series of RDT percentiles and coverage.
This figure plots descriptive statistics for RDT. Panel A shows cross-sectional percentiles of RDT over the sample. Panel B plots the monthly coverage of RDT relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDT. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on RDT-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.55]	0.52 [2.88]	0.62 [3.61]	0.67 [3.91]	0.80 [3.88]	0.32 [3.33]
α_{CAPM}	-0.12 [-1.96]	-0.06 [-0.88]	0.09 [1.25]	0.13 [2.07]	0.13 [2.18]	0.25 [2.64]
α_{FF3}	-0.17 [-2.87]	-0.12 [-2.01]	0.01 [0.15]	0.07 [1.30]	0.14 [2.27]	0.31 [3.30]
α_{FF4}	-0.20 [-3.38]	-0.13 [-2.12]	-0.04 [-0.61]	0.08 [1.49]	0.20 [3.25]	0.40 [4.28]
α_{FF5}	-0.22 [-3.70]	-0.22 [-3.64]	-0.15 [-2.79]	-0.06 [-1.20]	0.16 [2.55]	0.38 [4.02]
α_{FF6}	-0.25 [-4.02]	-0.22 [-3.59]	-0.18 [-3.24]	-0.04 [-0.80]	0.20 [3.33]	0.45 [4.76]
Panel B: Fama and French (2018) 6-factor model loadings for RDT-sorted portfolios						
β_{MKT}	0.99 [69.29]	0.99 [69.98]	0.96 [73.24]	0.96 [84.59]	1.06 [73.70]	0.07 [3.12]
β_{SMB}	0.11 [5.17]	0.03 [1.49]	-0.08 [-4.23]	-0.15 [-8.61]	-0.03 [-1.47]	-0.14 [-4.29]
β_{HML}	0.06 [2.34]	0.09 [3.53]	0.11 [4.43]	0.08 [3.62]	-0.00 [-0.17]	-0.07 [-1.63]
β_{RMW}	0.05 [1.94]	0.19 [7.07]	0.32 [12.57]	0.27 [12.14]	0.02 [0.66]	-0.04 [-0.83]
β_{CMA}	0.14 [3.49]	0.14 [3.41]	0.22 [5.93]	0.16 [4.90]	-0.08 [-1.93]	-0.22 [-3.52]
β_{UMD}	0.03 [2.23]	0.00 [0.01]	0.04 [2.95]	-0.03 [-2.43]	-0.07 [-4.80]	-0.10 [-4.56]
Panel C: Average number of firms (n) and market capitalization (me)						
n	431	406	355	309	349	
me (\$10 ⁶)	906	1116	1965	2517	2087	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RDT strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32 [3.33]	0.25 [2.64]	0.31 [3.30]	0.40 [4.28]	0.38 [4.02]	0.45 [4.76]
Quintile	NYSE	EW	0.28 [4.05]	0.27 [3.81]	0.22 [3.31]	0.23 [3.38]	0.12 [1.84]	0.13 [2.01]
Quintile	Name	VW	0.34 [3.61]	0.29 [3.08]	0.33 [3.58]	0.40 [4.24]	0.36 [3.82]	0.41 [4.32]
Quintile	Cap	VW	0.35 [3.39]	0.27 [2.62]	0.36 [3.67]	0.44 [4.48]	0.45 [4.58]	0.51 [5.16]
Decile	NYSE	VW	0.41 [3.12]	0.31 [2.38]	0.37 [2.90]	0.49 [3.85]	0.44 [3.36]	0.53 [4.10]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [2.99]	0.21 [2.24]	0.26 [2.77]	0.32 [3.42]	0.31 [3.32]	0.35 [3.80]
Quintile	NYSE	EW	0.13 [1.74]	0.13 [1.68]	0.07 [1.02]	0.08 [1.16]		
Quintile	Name	VW	0.31 [3.25]	0.25 [2.65]	0.28 [3.03]	0.32 [3.48]	0.30 [3.19]	0.33 [3.52]
Quintile	Cap	VW	0.32 [3.10]	0.23 [2.22]	0.31 [3.13]	0.35 [3.66]	0.38 [3.81]	0.41 [4.19]
Decile	NYSE	VW	0.37 [2.79]	0.25 [1.96]	0.31 [2.40]	0.38 [3.01]	0.35 [2.71]	0.41 [3.18]

Table 3: Conditional sort on size and RDT

This table presents results for conditional double sorts on size and RDT. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDT. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDT and short stocks with low RDT. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RDT Quintiles					RDT Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.65 [2.66]	0.69 [2.63]	0.85 [2.88]	0.73 [3.28]	0.87 [3.55]	0.22 [2.59]	0.22 [2.53]	0.18 [2.06]	0.21 [2.39]	0.15 [1.74]	0.18 [2.01]
	(2)	0.54 [2.23]	0.60 [2.61]	0.72 [3.10]	0.85 [3.94]	0.81 [3.39]	0.27 [2.55]	0.29 [2.74]	0.20 [1.98]	0.23 [2.18]	0.04 [0.37]	0.07 [0.69]
	(3)	0.66 [2.91]	0.61 [2.82]	0.80 [3.81]	0.82 [3.97]	0.87 [3.83]	0.21 [2.24]	0.21 [2.16]	0.15 [1.62]	0.22 [2.34]	0.11 [1.14]	0.17 [1.75]
	(4)	0.57 [2.75]	0.61 [3.04]	0.71 [3.49]	0.76 [3.85]	0.89 [4.14]	0.32 [3.50]	0.30 [3.26]	0.26 [2.79]	0.30 [3.29]	0.23 [2.55]	0.27 [2.95]
	(5)	0.45 [2.52]	0.53 [3.01]	0.55 [3.17]	0.64 [3.64]	0.77 [3.67]	0.32 [2.87]	0.22 [2.02]	0.30 [2.81]	0.40 [3.74]	0.39 [3.64]	0.47 [4.35]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RDT Quintiles					RDT Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	156	157	156	158	157	13	13	14	15	15	
	(2)	62	63	62	63	62	30	30	31	31	31	
	(3)	51	52	52	52	52	61	62	63	64	63	
	(4)	49	49	49	49	49	150	154	157	157	157	
(5)	50	50	51	50	51	998	1469	1630	1633	1550		

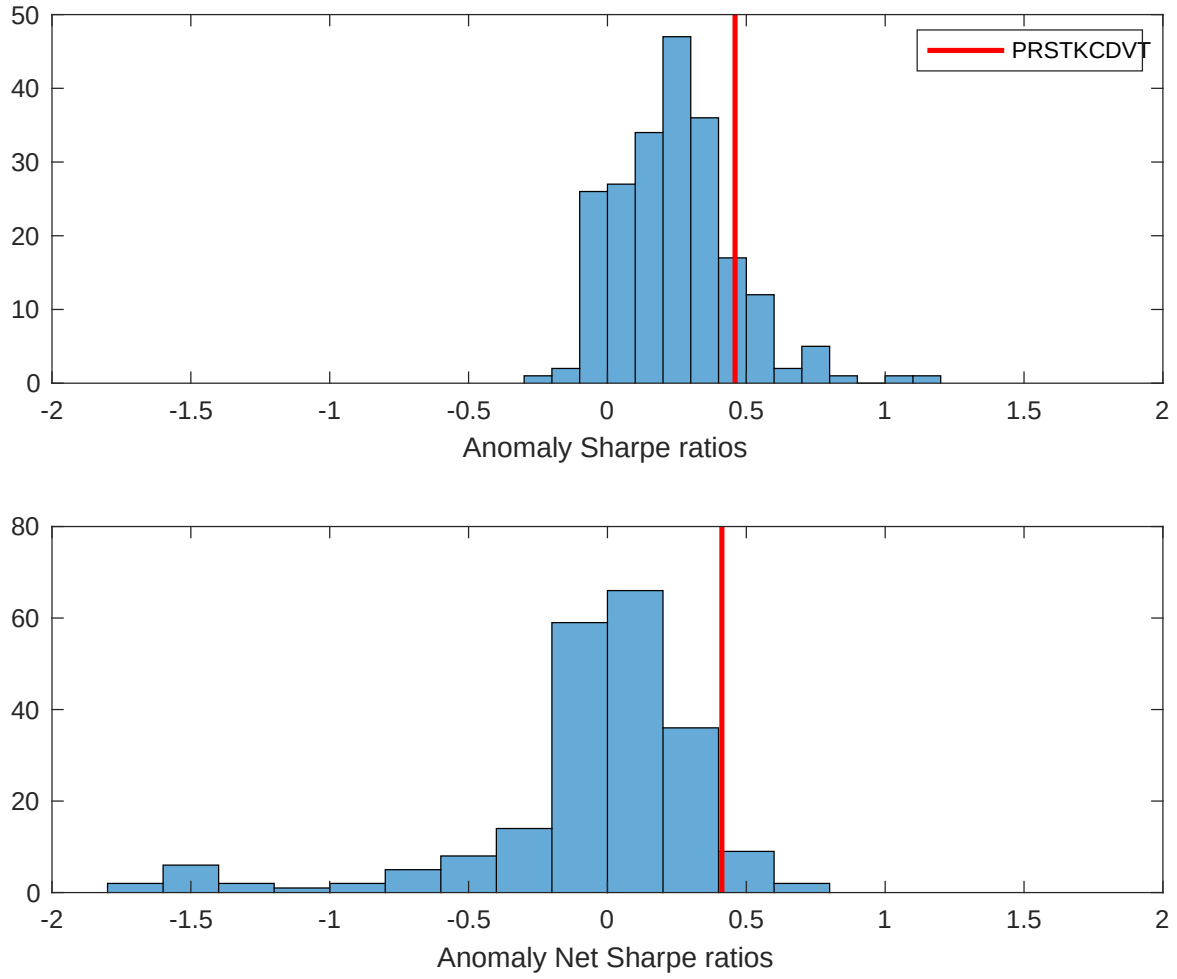


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDT with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

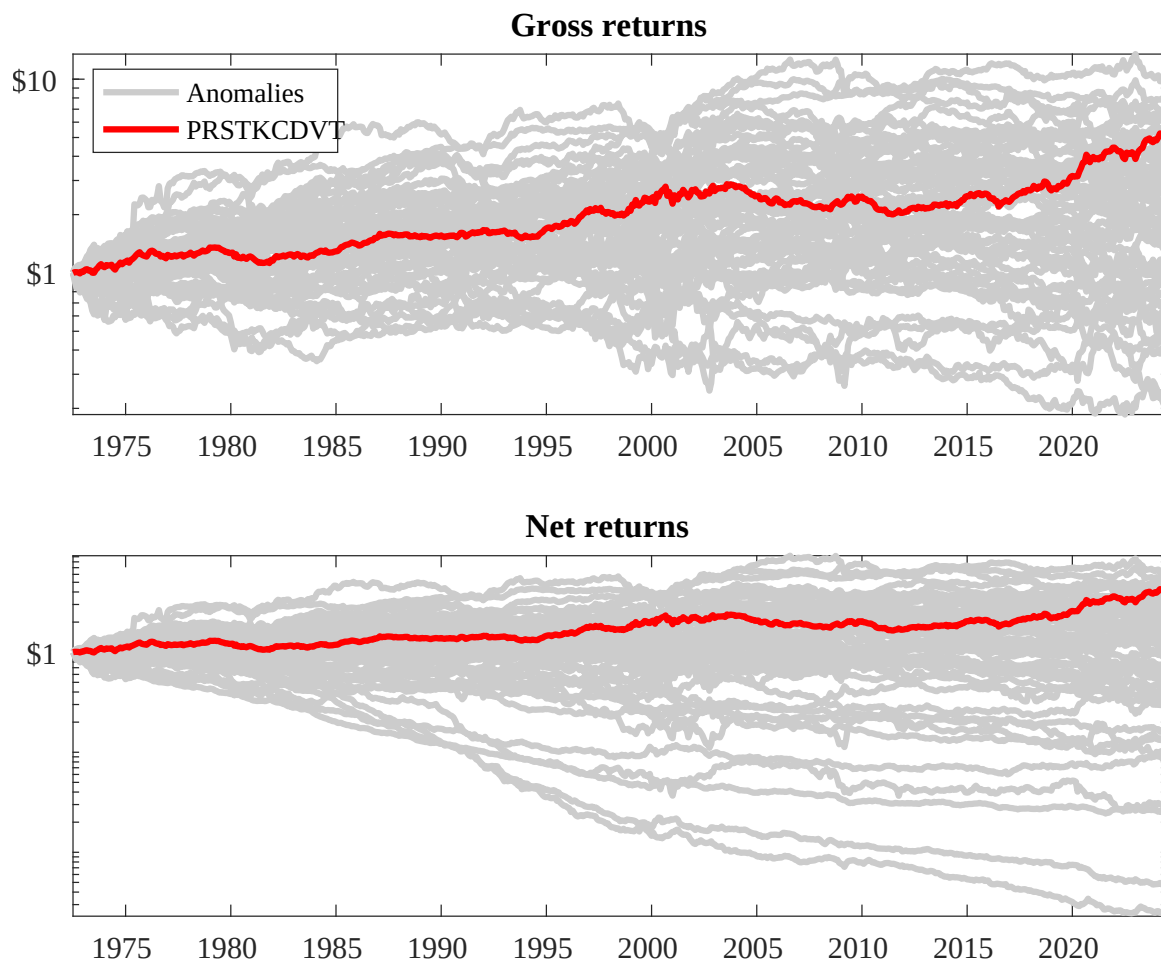
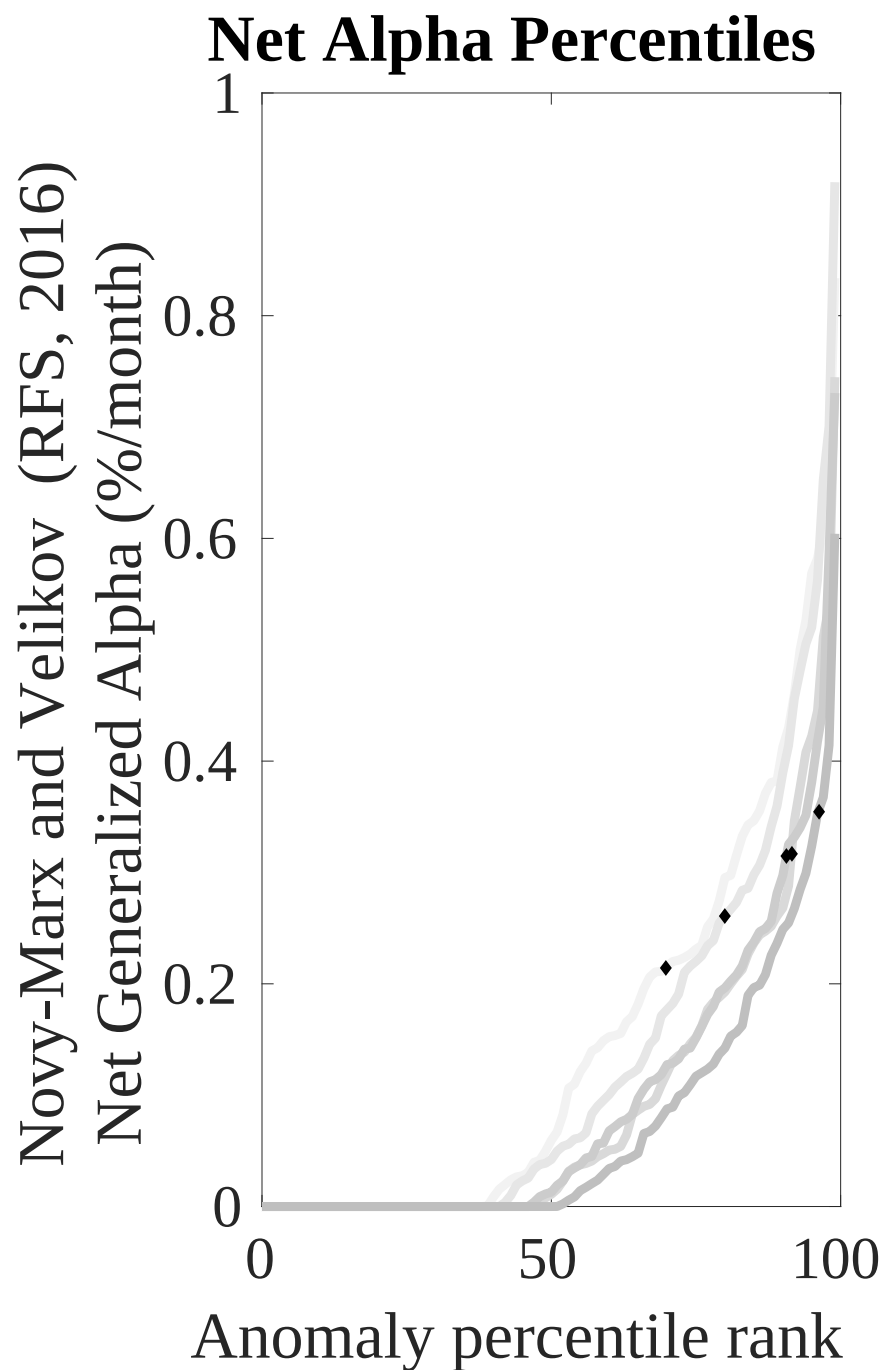
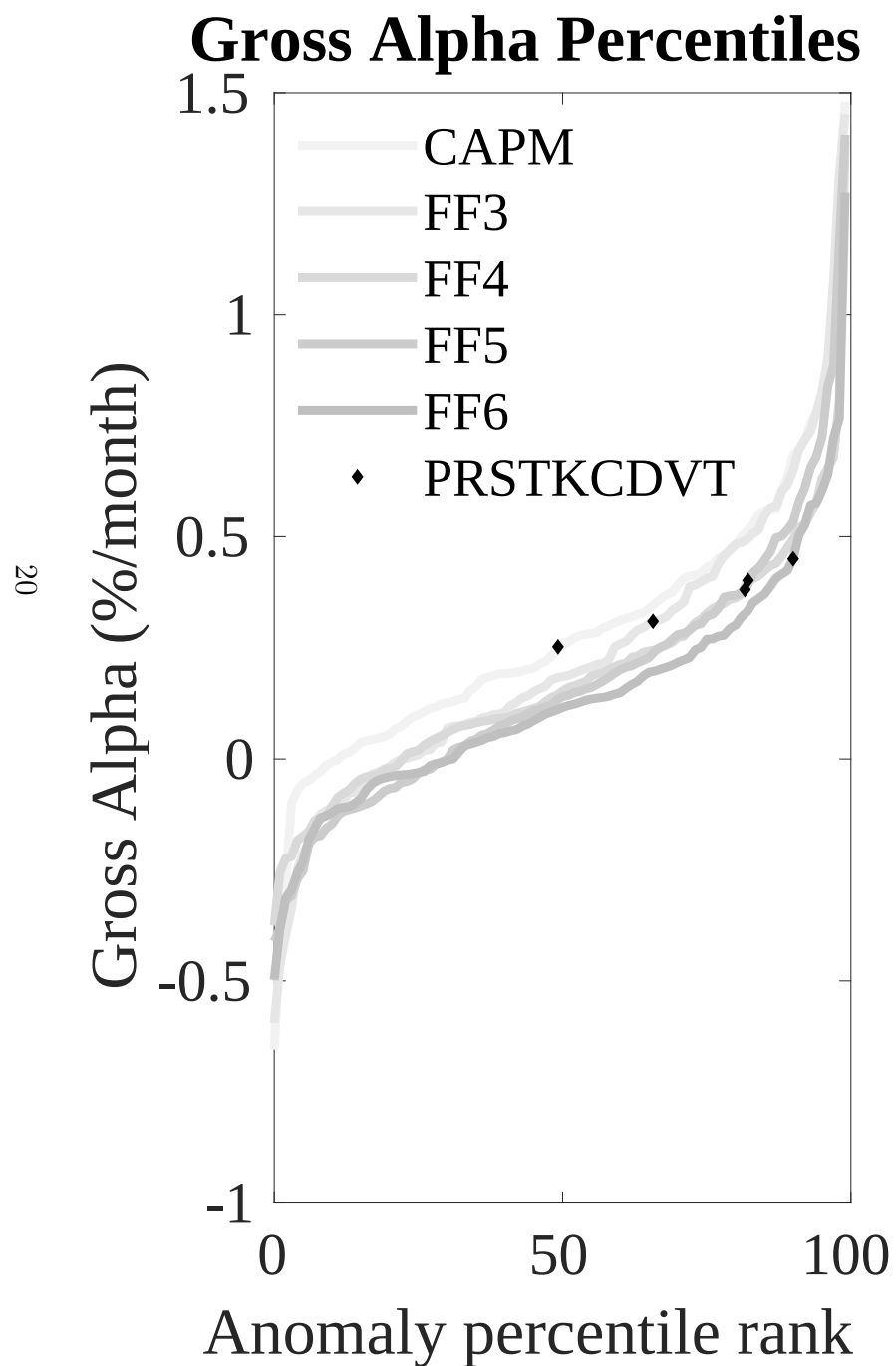


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDT trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



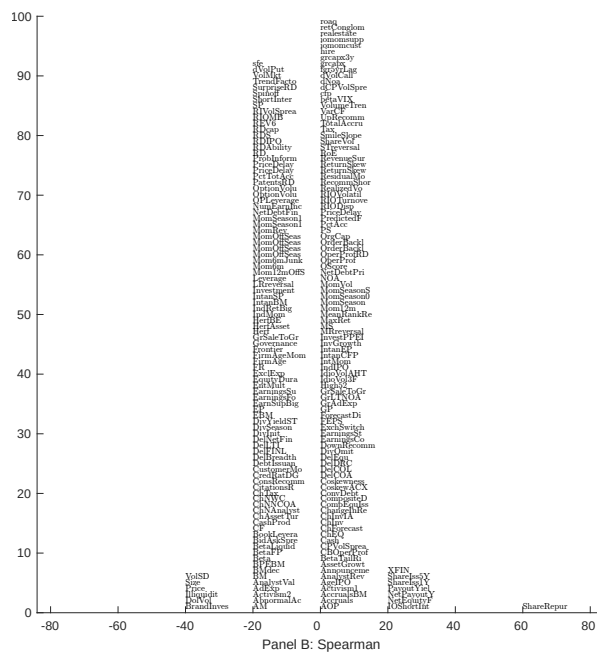
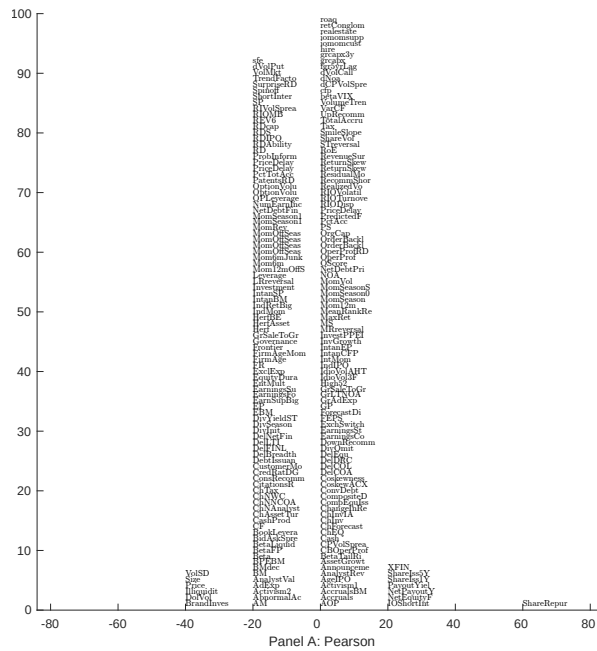


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with RDT. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

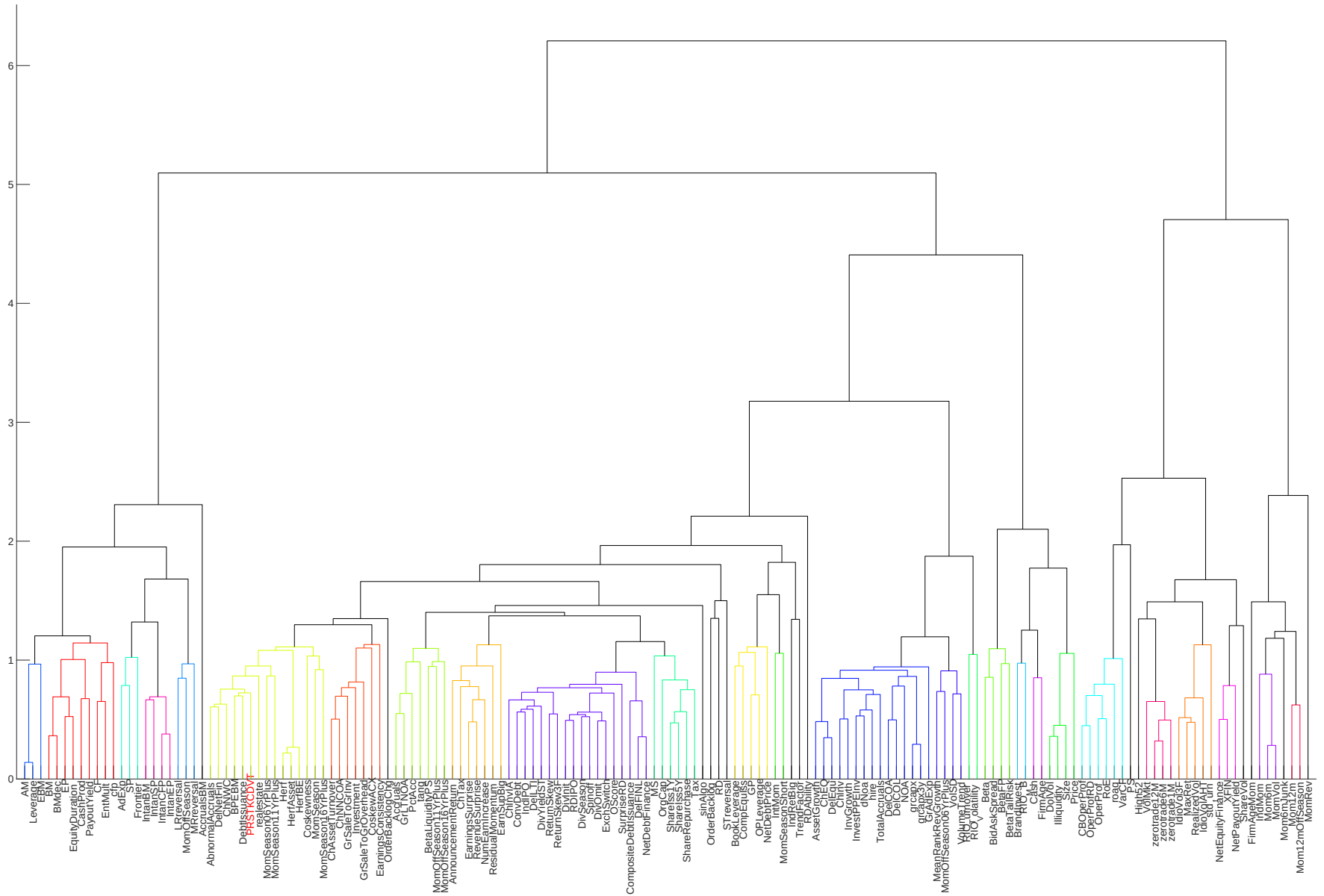


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

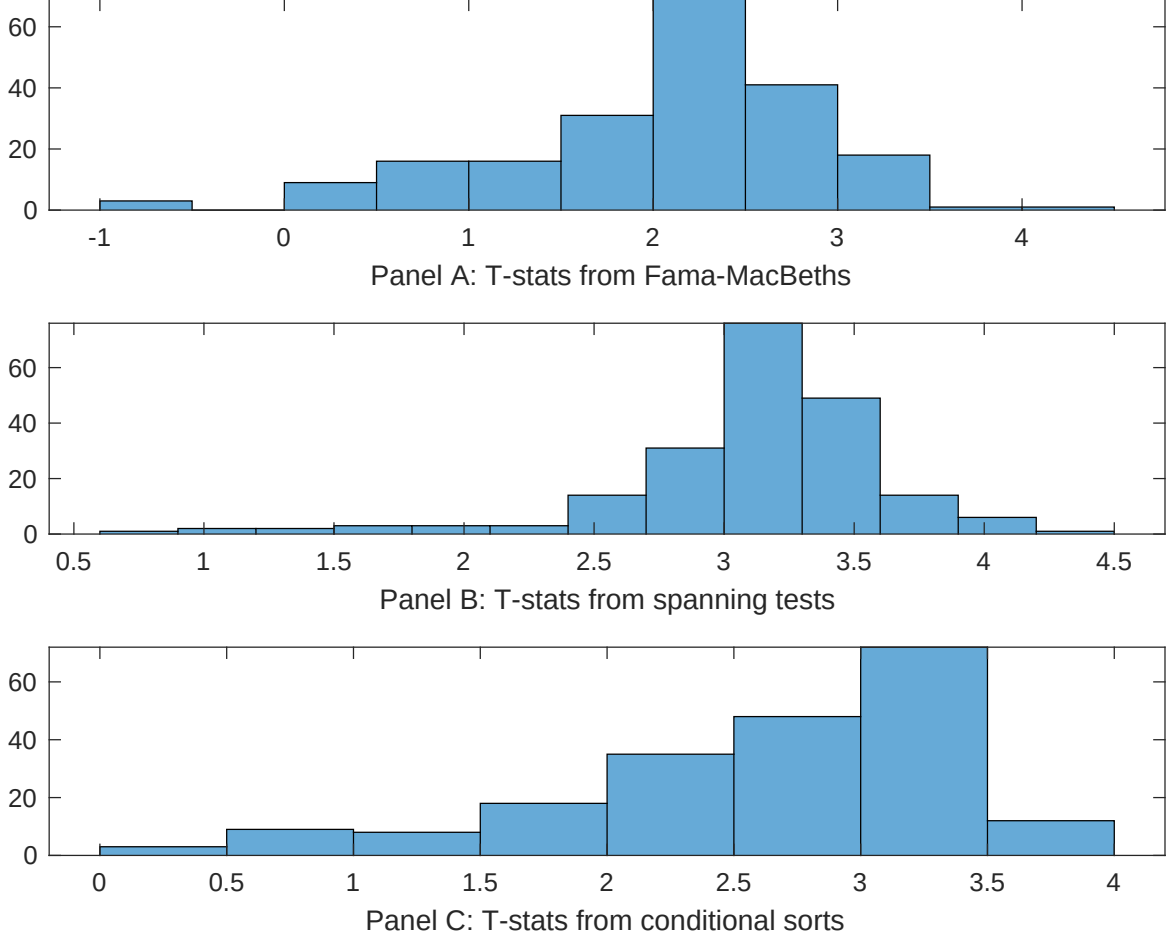


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RDT conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDT} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDT}RDT_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDT,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDT. Stocks are finally grouped into five RDT portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDT trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RDT. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDT}RDT_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.12 [5.86]	0.10 [4.52]	0.11 [4.83]	0.13 [7.30]	0.10 [4.80]	0.11 [4.89]	0.12 [6.16]
RDT	0.18 [2.46]	0.13 [1.85]	0.15 [2.02]	0.21 [2.71]	0.14 [1.83]	0.14 [1.88]	0.19 [2.81]
Anomaly 1	0.34 [2.66]						0.27 [2.84]
Anomaly 2		0.21 [1.09]					0.15 [0.09]
Anomaly 3			0.51 [1.67]				-0.77 [-0.93]
Anomaly 4				0.37 [3.06]			0.12 [0.90]
Anomaly 5					0.44 [0.53]		0.37 [1.93]
Anomaly 6						-0.11 [-0.74]	0.57 [1.24]
# months	619	614	619	619	619	619	614
$\bar{R}^2(\%)$	1	1	1	2	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RDT trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDT} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.44 [4.71]	0.37 [3.99]	0.41 [4.38]	0.37 [3.97]	0.42 [4.53]	0.41 [4.38]	0.29 [3.10]
Anomaly 1	-19.64 [-3.79]						-4.95 [-0.80]
Anomaly 2		-21.15 [-5.98]					-21.76 [-5.56]
Anomaly 3			-12.67 [-4.44]				3.16 [0.41]
Anomaly 4				-15.97 [-5.53]			-28.22 [-4.60]
Anomaly 5					-11.46 [-4.03]		9.28 [1.10]
Anomaly 6						-11.75 [-4.17]	6.57 [1.01]
mkt	2.76 [1.15]	2.12 [0.95]	1.37 [0.56]	-0.75 [-0.30]	1.78 [0.72]	1.58 [0.64]	-3.90 [-1.54]
smb	-1.59 [-0.36]	-0.46 [-0.12]	-13.40 [-4.15]	-21.50 [-6.06]	-14.70 [-4.51]	-16.72 [-4.99]	-8.89 [-1.64]
hml	-0.17 [-0.04]	-5.39 [-1.32]	-2.91 [-0.69]	-1.27 [-0.30]	-2.92 [-0.69]	-2.91 [-0.69]	-0.43 [-0.10]
rmw	3.00 [0.66]	0.61 [0.14]	4.27 [0.94]	5.71 [1.27]	4.20 [0.91]	4.17 [0.90]	6.10 [1.33]
cma	-20.62 [-3.28]	-13.60 [-2.15]	-15.28 [-2.38]	-14.87 [-2.35]	-16.25 [-2.53]	-16.59 [-2.60]	-10.07 [-1.59]
umd	-9.47 [-4.43]	-8.38 [-3.93]	-9.13 [-4.28]	-5.96 [-2.67]	-9.44 [-4.42]	-9.77 [-4.58]	-1.75 [-0.69]
# months	620	615	620	620	620	620	615
$\bar{R}^2(\%)$	16	18	16	18	16	16	22

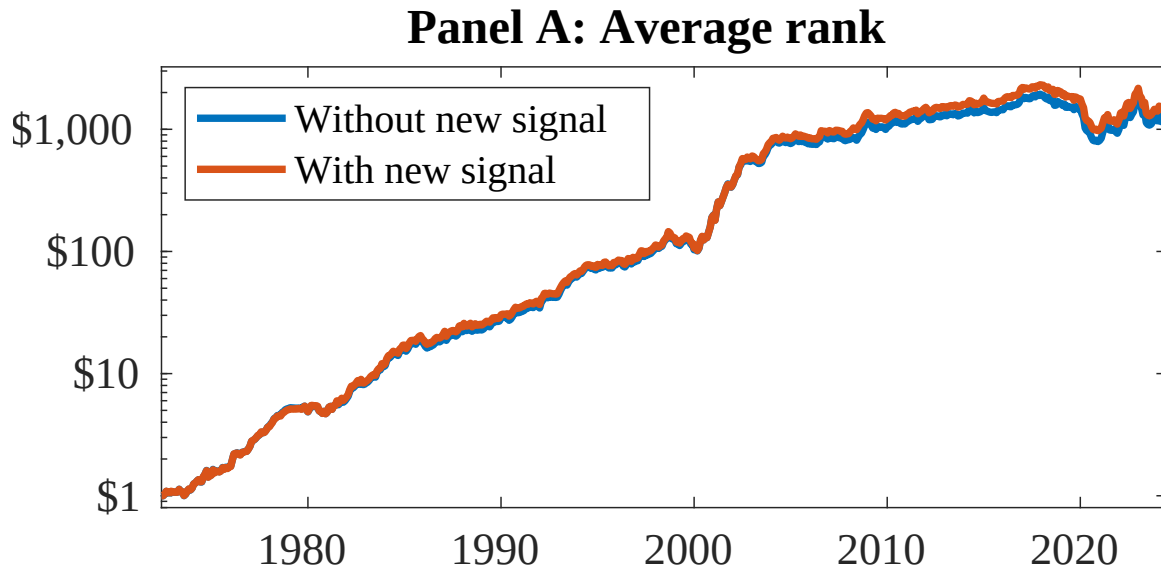


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as RDT. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Boudoukh, J., Michaely, R., Richardson, M., and Roberts, M. R. (2007). Cash distributions, aggregate payout, and the cross-section of returns. *Journal of Finance*, 62(5):2101–2137.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2001). Disappearing dividends: changing firm characteristics or lower propensity to pay? *Journal of Financial Economics*, 60(1):3–43.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Grullon, G. and Michaely, R. (2002). Dividends, share repurchases, and the substitution hypothesis. *Journal of Finance*, 57(4):1649–1684.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Parker, J. A. and Julliard, C. (2005). Consumption risk and the cross section of expected returns. *Journal of Political Economy*, 113(1):185–222.
- Pontiff, J. and Woodruff, F. (2023). Share repurchases and the cross-section of returns. *Journal of Finance*, 78(2):893–945.